PMSM Fault Diagnosis — Wavelet Scalogram + CNN

PMSM Fault Diagnosis using Wavelet Scalograms + CNN

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Research seminar — CNN classification of PMSM operating states from time–frequency images

1. The problem

- **PMSM** motors run EVs, robots, industrial servos, aircraft actuators.
- The most common stator fault — an **inter-turn short circuit** — can escalate to total winding failure within minutes.
- Goal: **detect faults automatically and early** from the motor's own signals.
- Classical methods (e.g. current spectrum analysis) need an expert to know *which* frequency to watch and break down when speed/load vary.

2. Our idea

Turn a **signal** problem into an **image** problem.

```
 signal → segment → CWT → scalogram image → CNN → fault class
(current/              (Morlet)  (224×224)       Healthy / InterTu
 vibration)                                  (+ Demag / Overload)
```

- The CNN **learns** the fault signatures itself — no hand-designed features.
- Two signal channels studied: **stator current** and **vibration**.

3. System pipeline

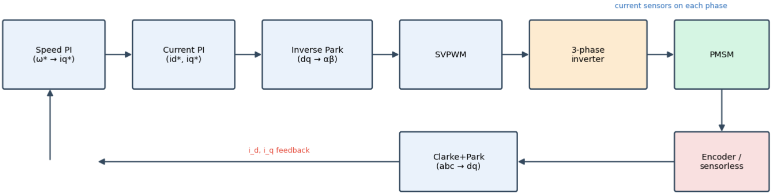
- **Sources:** real KAIST dataset (current + vibration), a synthetic generator, and MATLAB FOC/Simscape simulation.
- **One config + one manifest** drive everything (signal → scalogram → label → split): fully reproducible.
- Implemented in Python (TensorFlow/Keras, PyWavelets), MATLAB-free, **38 unit tests + CI**.

4. PMSM & its faults

- PMSM: permanent-magnet rotor locks onto the stator’s rotating field → rotates **synchronously**, no slip → high efficiency.
- Driven by **Field-Oriented Control (FOC)** — current sensors are already present.
- Faults studied:
  - **Inter-turn short** (primary, real data)
  - **Demagnetization**, **Overload** (synthetic)

Motor & control system

Fig. Field-Oriented Control (FOC) of a PMSM



- Stator (3-phase winding) + PM rotor; driven by an **inverter** under **FOC**.
- The current loop **rejects disturbances → it masks the fault in the current**.
- That is *why vibration detects inter-turn faults far better* (our key result).

PMSM vs its relatives

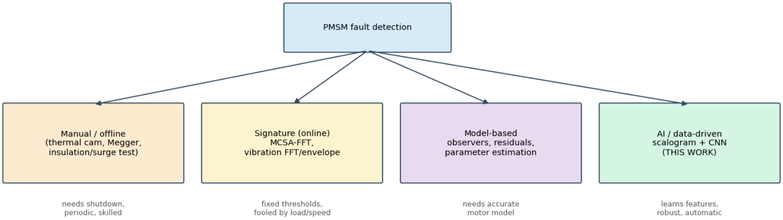
Fig. PMSM vs its relatives (PMDC / BLDC / Induction)

	PMSM	PMDC	BLDC	Induction
Commutation	Electronic (inverter)	Mechanical brushes	Electronic (inverter)	Electronic/DOL
Back-EMF	Sinusoidal	—(DC)	Trapezoidal	Sinusoidal
Efficiency	Very high	Medium	High	Medium
Torque density	Very high	Low	High	Medium
Control complexity	High (FOC)	Low	Medium	Medium/High
Maintenance	Low (no brushes)	High (brushes)	Low	Low
Rotor current	None	Yes (armature)	None	Induced (slip)
Typical use	EV, robotics, servo	Toys, simple drives	Drones, fans	Pumps, industry

- vs **PMDC**: same easy torque control, but no brushes → less maintenance.
- vs **BLDC**: sinusoidal back-EMF + FOC → smoother torque, higher efficiency.
- vs **Induction**: no rotor current/slip → higher efficiency & torque density.

Detecting the fault — today vs this work

Fig. Landscape of PMSM fault-detection methods



- Manual/offline (thermal, **Megger**, surge) → need shutdown, periodic.
- Online MCSA/vibration FFT → fixed thresholds, fooled by load/speed.
- **CWT scalogram + CNN** → learns features, robust, detects **earlier**.

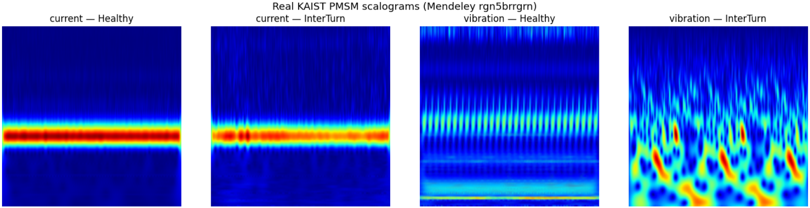
5. Why wavelets, not Fourier

- Fourier is **blind to time** — it can't say *when* a frequency occurs.
- Motor fault signals are **non-stationary** (transient, load-dependent).
- Uncertainty principle: $\Delta t \cdot \Delta f \geq \mathrm{const}$ — can't have perfect time *and* frequency resolution.
- **Wavelets** = short, localised waves → frequency-dependent resolution, ideal for transients.

6. The CWT & scalogram

- Scale the **Morlet** wavelet (frequency knob) and slide it (time knob).
- Each coefficient = **similarity** of the signal to that wave at that time.
- **Scalogram** = colour image of energy vs. time (x) and frequency (y).
- Bright bands & side-bands = fault signatures the CNN can *see*.

7. Example scalograms (real KAIST data)



Inter-turn fault visibly **enriches the vibration time–frequency content**.

8. The dataset

- **KAIST PMSM stator-fault dataset** (Mendeley `rgn5brrgrn`, CC-BY-4.0).
- Current @ **100 kHz**, vibration @ **25.6 kHz** → decimated to 10 kHz.
- **3,150** scalogram segments (0.5 s windows, 50 % overlap).

Channel	Healthy	InterTurn
current	200 (4 rec.)	1350 (27 rec.)
vibration	200 (4 rec.)	1400 (28 rec.)

9. Leakage-free split

- Split by **recording**, not by segment → no correlated windows across train/test (no data leakage).
- Stratified by class, **per channel**, 70 / 15 / 15.
- Special care: only **4 healthy recordings** → force ≥1 into each split so every class is evaluable.

10. Class imbalance — the key issue

- Only **4 healthy** vs ~60 fault recordings.
- Naïve training → model predicts “faulty” for everything: **80 % “accuracy” but 0 % healthy detection.**
- Fix: **undersample the majority in train/val**, keep **test natural**.
- Report **balanced accuracy** & **macro-F1**, not raw accuracy.

11. CNN architecture

```
224×224×3
 → Conv32 → Pool → Conv64 → Pool → Conv128 → Pool
 → Flatten → Dropout(0.5) → Dense128 → Softmax
```

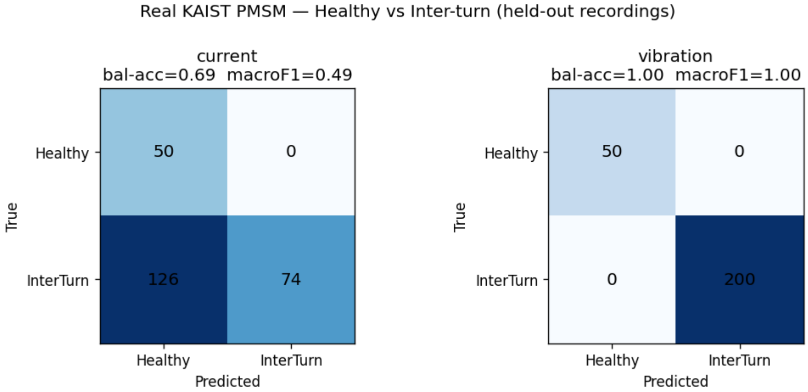
- Adam, sparse cross-entropy, early stopping, data augmentation.
- **Fusion** model: two branches (current + vibration) → global avg pool → concatenate → dense.

12. Results — real data

Channel	Balanced acc	Macro-F1	Healthy rec.	InterTurn re
Vibration	**1.00**	**1.00**	1.00	1.00
Fusion (cur.+vib.)	0.88	0.76	1.00	0.75
Current	0.69	0.49	1.00	0.37

Held-out recordings, Healthy vs Inter-turn.

13. Confusion matrices



Vibration is perfect; current misses most faults; fusion is in between.

14. Analysis

- **Vibration $\gg$ current** for inter-turn detection — physically expected (vibration responds directly; FOC suppresses current signatures).
- **Fusion** beats current but not vibration alone on this small data.
- Right **metric matters**: balanced accuracy exposed the majority-class collapse that raw accuracy hid.

15. Improvement experiments

- **Balancing** removes the majority-class collapse (current healthy recall 0.00 → 1.00).
- **Image size** & **training-set size** studied (see `docs/experiments.md`).
- Overfitting controlled by dropout, global average pooling, augmentation and early stopping.

16. Limitations

- Only **4 distinct healthy recordings** → perfect vibration score can't yet exclude a *recording-identity* shortcut.
- Demagnetization / Overload only synthetic.
- Fusion comparison indicative, not strictly controlled.

17. Conclusions & future work

- A reproducible scalogram + CNN pipeline detects inter-turn PMSM faults; **vibration reaches balanced accuracy 1.00** on held-out recordings.
- Future: more independent recordings; paired multi-channel fusion; severity-aware targets; explainability (Grad-CAM); on-line deployment in the drive.

Thank you

Questions?

Code, data instructions and full report:
`github.com/molhamfetnah/pmsm-fault-diagnosis-cnn-scalogram`